

Flight Information System using HBase & HiveQL

◆ Step 1: Start HBase and Hive

```
Bash
start-hbase.sh
hive
```

◆ a. Create Flight Info HBase Table

```
Bash
create 'flight_info', 'info', 'schedule', 'delay'
```

Insert Data

```
Bash
put 'flight_info', 'F1', 'info:name', 'Indigo'
put 'flight_info', 'F1', 'schedule:time', '10AM'
put 'flight_info', 'F1', 'delay:departure', '15'

put 'flight_info', 'F2', 'info:name', 'AirIndia'
put 'flight_info', 'F2', 'schedule:time', '2PM'
put 'flight_info', 'F2', 'delay:departure', '5'
```

Display Data

```
Bash
scan 'flight_info'
```

◆ b. Create, Alter, Drop Table in HBase

Create Table

```
Bash
create 'test', 'cf1'
```

Alter Table

```
Bash
alter 'test', 'cf2'
```

Drop Table

```
Bash
disable 'test'
drop 'test'
```

◆ c. Create External Hive Table

Open Hive:

```
Bash
hive
```

Hive Table Mapping

```
SQL
CREATE EXTERNAL TABLE hive_flight(
flight_id STRING,
name STRING,
time STRING,
departure_delay INT
)

STORED BY 'org.apache.hadoop.hive.hbase.HBaseStorageHandler'

WITH SERDEPROPERTIES (
"hbase.columns.mapping"=":key,info:name,schedule:time,delay:departure"
)

TBLPROPERTIES ("hbase.table.name"="flight_info");
```

◆ d. Total Departure Delay

```
SQL
SELECT SUM(departure_delay) FROM hive_flight;
```

◆ e. Average Departure Delay

```
SQL
SELECT AVG(departure_delay) FROM hive_flight;
```

◆ f. Create Index on Flight Table

```
SQL
CREATE INDEX idx_name
ON TABLE hive_flight(name)
AS 'COMPACT'
WITH DEFERRED REBUILD;
```

Rebuild Index

```
SQL
ALTER INDEX idx_name ON hive_flight REBUILD;
```



Important Commands

Show HBase Tables

```
Bash  
list
```

Describe Table

```
Bash  
describe 'flight_info'
```

Exit HBase

```
Bash  
exit
```

Exit Hive

```
Bash  
quit;  
■
```